

Finding Pareto Stationary Point of MILP by BCD/ADMM

1. Optimization problem we want to solve

$$\begin{aligned} \min_{\mathbf{x}, \mathbf{z}} \quad & \mathbf{1}^\top G \mathbf{z} \\ \text{s.t.} \quad & A\mathbf{x} \leq b, \quad \mathbf{x} \in \{0, 1\}^N \\ & \mathbf{z} \geq E\mathbf{x} + \mathbf{f} - \mathbf{1} \\ & \mathbf{z} \geq \mathbf{0} \end{aligned}$$

1.1 Major characteristics of given MILP problem

1.

$$\mathbf{x}_s = \begin{bmatrix} \mathbf{x}_{s_1} \\ \mathbf{x}_{s_2} \\ \vdots \\ \mathbf{x}_{s_p} \end{bmatrix} \rightarrow \mathbf{x} = P\mathbf{x}_s$$

where P is a permutation matrix (given), or index mapping (computable).

2. Decision variables for each agent:

$$\mathbf{x}_{s_i} = \{\mathbf{x}_{s_1}, \mathbf{x}_{s_2}, \dots, \mathbf{x}_{s_p}\}$$

Feasibility of each agent's decision variables are decoupled, i.e.,

$$A_{s_1}\mathbf{x}_{s_1} \leq b_{s_1}, \quad A_{s_2}\mathbf{x}_{s_2} \leq b_{s_2}, \quad \dots, \quad A_{s_p}\mathbf{x}_{s_p} \leq b_{s_p}.$$

Therefore, the value of $\mathbf{x}_{s_i}$ cannot change the feasible region of other $\mathbf{x}_{s_j}$.

3. The feasible region for each element of $\mathbf{z}$ will be classified into two cases:

$$z_i \geq 0 \quad \text{or} \quad z_i \geq 1.$$

4. Each agent's decision can influence the feasible region of $\mathbf{z}$.

Example 1 (one agent changes decision while other's decision remains constant):

$\mathbf{x}'_{s_1}$ changes and $\mathbf{x}_{s_2}, \dots, \mathbf{x}_{s_p}$ remains constant,

$$\mathbf{x}' = P \begin{bmatrix} \mathbf{x}'_{s_1} \\ \mathbf{x}_{s_2} \\ \vdots \\ \mathbf{x}_{s_p} \end{bmatrix} \Rightarrow \mathbf{z}' \leq E\mathbf{x}' + \mathbf{f} - \mathbf{1} = EP \begin{bmatrix} \mathbf{x}'_{s_1} \\ \mathbf{x}_{s_2} \\ \vdots \\ \mathbf{x}_{s_p} \end{bmatrix} + \mathbf{f} - \mathbf{1}$$

Example 2 (every agent changes their decision simultaneously):

Every agent $\mathbf{x}'_{s_1}, \mathbf{x}'_{s_2}, \dots, \mathbf{x}'_{s_p}$ changes simultaneously,

$$\mathbf{x}' = P \begin{bmatrix} \mathbf{x}'_{s_1} \\ \mathbf{x}'_{s_2} \\ \vdots \\ \mathbf{x}'_{s_p} \end{bmatrix} \Rightarrow \mathbf{z}' \leq E\mathbf{x}' + \mathbf{f} - \mathbf{1}$$

Changing $\mathbf{x}_{s_i} \rightarrow \mathbf{x}'_{s_i}$ does not influence the feasibility (or doable) of changing $\mathbf{x}_{s_j} \rightarrow \mathbf{x}'_{s_j}$, but multiple simultaneous changes may alter the feasible region of the same element of $\mathbf{z}$.

Therefore, we can consider

- sequential decision change
- parallel decision change

5. G is a diagonal matrix with all positive diagonal values.

- If the feasible region of z_i satisfies $z_i \geq 0$ or $z_i \geq 1$, then corresponding z_i will be immediately decided to $z_i = 0$ and $z_i = 1$.
- Therefore, G functions as a weighting matrix, and no additional computation cost arises if the feasible space of $\mathbf{z}$ is defined.

2. Multi-Task Learning Strategy to Solve Given MILP

- BCD (Block Coordinate Descent): Sequential / Stackelberg game approach.
- ADMM (Alternating Direction Method of Multipliers): Parallel / dynamic game approach for task consensus.

2.1 BCD Approach

Each agent s_i decides optimal $\mathbf{x}_{s_i}$ while assuming other agents' decisions $\mathbf{x}_{s_{-i}}$ remain constant. The cost function should be decreased if the agent changes its decision.

2.1.1 BCD Process

1. Input:

$$\mathbf{x}^k = P \begin{bmatrix} \mathbf{x}_{s_1}^k \\ \vdots \\ \mathbf{x}_{s_p}^k \end{bmatrix}, \quad \mathbf{z}^k$$

2. Iteration: solve the following MILP for each agent sequentially

$$\begin{aligned} \mathbf{x}_{s_i}^{k+1} &= \arg \min_{\mathbf{x}_{s_i}} \mathbf{1}^\top G \mathbf{z} \\ \text{s.t. } A_{s_i} \mathbf{x}_{s_i} &\leq b_{s_i}, \quad \mathbf{z} \geq EP \begin{bmatrix} \mathbf{x}_{s_1}^k \\ \vdots \\ \mathbf{x}_{s_i}^k \\ \vdots \\ \mathbf{x}_{s_p}^k \end{bmatrix} + \mathbf{f} - \mathbf{1} \end{aligned}$$

3. Continue until $\mathbf{x}_{s_i}^{k+1} = \mathbf{x}_{s_i}^k$ for all s_i .

2.2 ADMM with Heuristic/Surrogate Approach

Each agent computes its optimal decision $\mathbf{x}_{s_i}$ and corresponding $\mathbf{z}_{s_i}$, assuming other agents' decisions remain constant. The goal is to enforce consensus:

$$\mathbf{z}_{s_1} = \mathbf{z}_{s_2} = \cdots = \mathbf{z}_{s_p} = \mathbf{z}$$

2.2.1 ADMM Consensus Optimization Problem

$$\begin{aligned} \min_{\mathbf{z}_{s_i}, \mathbf{z}} \quad & \sum_{i=1}^p \mathbf{1}^\top G \mathbf{z}_{s_i} \\ \text{s.t.} \quad & \mathbf{z}_{s_i} = \mathbf{z}, \quad \forall i = 1, \dots, p. \end{aligned}$$

Augmented Lagrangian for each agent:

$$\mathcal{L}_\rho = \sum_{i=1}^p \left[\mathbf{1}^\top G \mathbf{z}_{s_i} + \boldsymbol{\lambda}_{s_i}^\top (\mathbf{z}_{s_i} - \mathbf{z}) + \frac{\rho}{2} \|\mathbf{z}_{s_i} - \mathbf{z}\|_2^2 \right]$$

2.2.2 ADMM Consensus Process

1. $\mathbf{x}_{s_i}, \mathbf{z}_{s_i}$ update: for each s_i ,

$$(\mathbf{x}_{s_i}^{k+1}, \mathbf{z}_{s_i}^{k+1}) = \arg \min_{\mathbf{x}_{s_i}, \mathbf{z}_{s_i}} \mathbf{1}^\top G \mathbf{z}_{s_i} + \boldsymbol{\lambda}_{s_i}^\top (\mathbf{z}_{s_i} - \mathbf{z}^k) + \frac{\rho}{2} \|\mathbf{z}_{s_i} - \mathbf{z}^k\|_2^2$$

$$\text{s.t. } A_{s_i} \mathbf{x}_{s_i} \leq b_{s_i}, \quad \mathbf{z}_{s_i} \geq EP \begin{bmatrix} \mathbf{x}_{s_1}^k \\ \vdots \\ \mathbf{x}_{s_i}^k \\ \vdots \\ \mathbf{x}_{s_p}^k \end{bmatrix} + \mathbf{f} - \mathbf{1}, \quad \mathbf{z}_{s_i} \geq \mathbf{0}$$

2. $\mathbf{x}$ update:

$$\mathbf{x}^{k+1} = P \begin{bmatrix} \mathbf{x}_{s_1}^{k+1} \\ \vdots \\ \mathbf{x}_{s_p}^{k+1} \end{bmatrix}$$

3. $\mathbf{z}$ update (two options):

Option (i):

$$\mathbf{z}^{k+1} = \max \mathbf{z}_{s_i}^{k+1} \quad (\text{element-wise})$$

Option (ii):

$$\begin{aligned} \mathbf{z}^{k+1} &= \arg \min_{\mathbf{z}} \sum_{i=1}^p \left[\boldsymbol{\lambda}_{s_i}^\top (\mathbf{z}_{s_i}^{k+1} - \mathbf{z}) + \frac{\rho}{2} \|\mathbf{z}_{s_i}^{k+1} - \mathbf{z}\|_2^2 \right] \\ \text{s.t. } \mathbf{z} &\geq E \mathbf{x}^{k+1} + \mathbf{f} - \mathbf{1}, \quad \mathbf{z} \geq \mathbf{0} \end{aligned}$$

4. Dual update: for each s_i ,

$$\boldsymbol{\lambda}_{s_i}^{k+1} = \boldsymbol{\lambda}_{s_i}^k + \rho(\mathbf{z}_{s_i}^{k+1} - \mathbf{z}^{k+1})$$

Continue until $\mathbf{z}_{s_i} = \mathbf{z}$ for all s_i . If iteration stops, it reached to Pareto stationary.

2.2.3 Global z -update (in ADMM-Option (ii))

$$\begin{aligned} z^{k+1} &= \arg \min_z \sum_{i=1}^p \left[(\lambda_i^k)^\top (z_{s_i}^{k+1} - z) + \frac{\rho}{2} \|z_{s_i}^{k+1} - z\|_2^2 \right] \\ \text{s.t. } z &\geq EP \mathbf{x}^{k+1} + \mathbf{f} - \mathbf{1}, \quad z \geq \mathbf{0} \end{aligned}$$

1. Quadratic form (dropping constants).

$$z^{k+1} = \arg \min_z \frac{\rho p}{2} z^\top z - \left(\sum_{i=1}^p \lambda_i^k + \rho \sum_{i=1}^p z_{s_i}^{k+1} \right)^\top z$$

2. Unconstrained minimizer

$$z_{\text{unproj}}^{k+1} = \frac{1}{p} \sum_{i=1}^p z_{s_i}^{k+1} + \frac{1}{\rho p} \sum_{i=1}^p \lambda_i^k$$

Define averages,

$$\begin{aligned} \bar{z}^{k+1} &:= \frac{1}{p} \sum_{i=1}^p z_{s_i}^{k+1} \\ \bar{\lambda}^k &:= \frac{1}{p} \sum_{i=1}^p \lambda_i^k \end{aligned}$$

Then,

$$z_{\text{unproj}}^{k+1} = \bar{z}^{k+1} + \frac{1}{\rho} \bar{\lambda}^k$$

3. Projection onto the feasibility set

$$\begin{aligned} \ell^{k+1} &= \max(EP \mathbf{x}^{k+1} + \mathbf{f} - \mathbf{1}, \mathbf{0}) \Rightarrow \text{lowerbound} \\ z^{k+1} &= \Pi_{z \geq \ell^{k+1}}(z_{\text{unproj}}^{k+1}) \end{aligned}$$

4. Final closed-form update (element-wise)

$$\boxed{z^{k+1} = \max\left(\bar{z}^{k+1} + \frac{1}{\rho} \bar{\lambda}^k, EP \mathbf{x}^{k+1} + \mathbf{f} - \mathbf{1}, \mathbf{0}\right)}$$